Title: Incidence of acute hamstring injuries in soccer: A systematic review of 13 studies involving more than 3800 athletes with 2 million sport exposure hours

Authors: W.M.Diemer, M.Winters, J.L.Tol, H.I.M.F.L.Pas, M.H.Moen

Affiliations

Willemijn M. Diemer, MD

Department of sports medicine, Meander Medical Center, Amersfoort/Baarn, The Netherlands

Marinus Winters, PhD, MSc, PT

Research Unit for General Practice in Aalborg, Department of Clinical Medicine, Aalborg University, Aalborg, Denmark.

Johannes L. Tol, PhD, MD

Amsterdam University Medical Centers, Academic Medical Center, Amsterdam, The Netherlands

Amsterdam Movement Sciences, Academic Center for Evidence Based Medicine, Amsterdam IOC Center, ACHSS, The Netherlands

Haiko I.M.F.L.Pas, MD

Amsterdam University Medical Centers, Academic Medical Center, Amsterdam

Amsterdam Movement Sciences, Academic Center for Evidence Based Medicine, Amsterdam IOC Center, ACHSS, The Netherlands

Northwest clinics, department of pulmunology; Almaar; the Netherlands

Maarten H. Moen, PhD, MD

Sports Medicine Department, Onze Lieve Vrouwe Gasthuis, Amsterdam, The Netherlands.

Bergman Clinics, Naarden, The Netherlands

Conflicts of interest and ethical approval

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The authors report no conflicts of interest. No funds or sources of financial assistance were used to conduct this review, or used to assist with the preparation of the manuscript. As a systematic review, ethical approval was not sought in line with our institutional guidance.

Registration

PROSPERO, the International Prospective Register of Systematic Reviews (CRD42016039011).

Corresponding author

Willemijn. M. Diemer

Department of sports medicine, Meander Medical Center, Maatweg 3, 3813 T.Z. Amersfoort, The Netherlands

Email: wmdiemer@gmail.com

Phone number: 0031644158103

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Abstract

Objective To estimate the incidence and recurrence rate of acute hamstring injuries in all levels of soccer.

Study design Epidemiology systematic review

Literature search We searched the PubMed (including Medline), CINAHL, SPORTDiscus, Embase and CENTRAL electronic databases.

Study selection criteria We included prospective studies of all levels of adult soccer players that registered acute hamstring injuries and provided a description of incidence of acute hamstring injuries in n/1000 playing hours (or available data to calculate this).

Data synthesis Due to heterogeneity, we synthesised the data descriptively.

Results Thirteen studies including 3868 players met the inclusion criteria. Two of thirteen included studies reported on hamstring injuries in women and thirteen on men. The incidence of acute hamstring injury ranged from 0.3 to 0.5 per 1000 exposure hours in women and 0.3 to 1.9 per 1000 exposure hours in men. Hamstring injuries accounted for 5% to 15% of all soccer-related injuries. Hamstring injury recurrence rates ranged from 4% to 68%, depending on the injury definition. Certainty of the evidence ranged from moderate to very low.

Conclusion The incidence of acute hamstring injury in soccer was 0.3 to 1.9 per 1000 exposure hours. The recurrence rate was 4-68%. Certainty of the evidence ranged from moderate to very low.

Key Words: Epidemiology/survey research, Lower extremity, Prevention (injury), Soccer, Systematic review/meta-analysis

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Introduction

Soccer (football) has over 265 million active players worldwide.[11] Healthy lifestyle, positive influence on development and education of children and economic advantages are just some of the benefits from participation in soccer.[7,8,13,24] Despite its positive effects, the risk of injuries in matches (i.e. game) and practice (training) is relatively high. Soccer accounts for the largest number of hospital-treated sports injuries in the European Union (>1.5 million visits).[2]

An acute hamstring injury is frequently reported as one of the most common soccer injuries,[3,10] but the incidence of acute hamstring injuries in soccer is unclear. Accurate information about the incidence of acute hamstring injuries helps quantify the injury burden, and guides resource allocation for hamstring injury prevention. Understanding injury incidence may also help team doctors and physical therapists to make better decisions in targeting strategies for the prevention and rehabilitation of soccer injuries.

The purpose of this systematic review was to estimate the incidence and recurrence rate of acute hamstring injuries in soccer.

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Methods

This review was prospectively registered in PROSPERO, the International Prospective Register of Systematic Reviews (CRD42016039011). The Preferred Items for Systematic Reviews and Meta-Analysis (PRISMA) statement were followed in preparing, conducting and reporting this systematic review.[25]

Research question and outcomes

Our primary research aim was to estimate the incidence of acute hamstring (posterior thigh) injuries (indirect structural muscle injury[26]) during training and matches in adult male and female soccer of all playing levels.

The primary outcomes were: 1) incidence rate in n/1000 playing hours during training and match play, 2) incidence proportion, i.e. the proportion of hamstring injuries out of the total injuries in soccer.

Our secondary aim was to estimate the recurrence rate of acute hamstring injuries[12] during training and match play in adult male and female soccer of all playing levels as a proportion of the total injuries in soccer.

Acute hamstring injuries were defined as any suddenly occurring, indirect, structural muscle injury to the hamstrings or the posterior thigh, sustained during soccer match or training, causing tissue injury, time loss or the need for medical assistance.

Inclusion criteria for studies

Studies were required to meet the following criteria: (1) prospective design, (2) adult soccer players (age ≥18 years), (3) soccer-specific population (11-a-side), all levels of competition, (4) full text report available, (5) description of at least one primary outcome, (6) acute hamstring injuries, defined as acute onset, acute injuries to the posterior thigh, presentation within 7 days of injury, and (7)

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hamstring injury causing either: tissue injury[12,17], time loss[12], or the need for medical assistance[12]. We used previously defined, well-accepted definitions of acute hamstring injury.[12,17] (Box 1)

Literature search

A sensitive search strategy was constructed and executed with the assistance of a research librarian. The search strategy consisted of indexed and free text terms. Examples of search terms used were: 'hamstring', 'thigh', 'semitendinosus', 'semimembranosus', 'biceps femoris', 'injury', 'rupture', 'sprain', 'strain', 'tear', 'epidemiology', 'incidence', 'occurrence', 'soccer', 'football' (full search available in supplementary appendix 1). Electronic databases (PubMed (including MEDLINE), CINAHL, SPORTDiscus, Embase and Cochrane) were searched by one investigator (WD) for eligible studies without language or time restriction, from their inception until August 6th 2019. All references of the primarily selected articles and recent systematic reviews were searched manually for potentially relevant articles.

Study selection

All search results were downloaded into an electronic citation manager (Mendeley 1.15.3, 2008-2015 Mendeley Ltd.) and duplicates were removed. When studies had an overlap in player population, the study with the largest population was included. All remaining titles and abstracts were independently screened by two reviewers (WD, MM). Full text articles were evaluated for inclusion independently by two authors (WD, MM). In case of disagreement, consensus was sought through discussion among the two authors. If no consensus was reached, a third author (JT) took the final decision.

Data extraction

Two authors (WD, HP) extracted data from included studies using a data extraction form adapted from the Cochrane Collaboration.[15] Study authors were contacted in case of missing information. For

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randomised controlled trials, only data from the control (i.e. not receiving intervention) group were extracted.

We extracted: study design, sample size, follow up time, the number of recurrent hamstring injuries[12], definition of hamstring/posterior thigh injury, mean age, sex, competition level (amateur, professional) and primary outcomes (I. Incidence in n/1000 playing hours. II. proportion of hamstring injuries out of the total number of injuries). All numeric data were extracted. Where calculations were made to derive data, a third author (MW) checked the calculations.

Quality assessment

Two authors assessed the study quality (WD, HP) using a modified Newcastle Ottawa Scale for cohort studies with a seven-star maximum (see Appendix 2).[15] Modifications included adding an observation domain and more focus on injury registration. The following four domains were appraised for quality of reporting:

1. Selection: one star for description of population; one star for recruitment of population, and one star for description of baseline health and injury status.
2. Observation: one star for surveillance methods and one star when time of observation was clear.
3. Outcome: one star for accurate and clear definition of acute hamstring injuries or "acute injury to the posterior thigh".
4. Other: one star for appropriate other methods.

If no consensus was reached following discussion, a third author (MM) made the final decision.

Data synthesis

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Quantitative analysis was precluded due to methodological heterogeneity in the definition of soccer season, the start of the study in the calendar year and duration of data collection. Instead, we conducted a descriptive synthesis (see Table 1).

Hamstring incidence was synthesized according to the following subgroups; injury definition, level of play and sex. We planned grouping findings according to study quality.

Certainty of the evidence (GRADE)

The Grading of Recommendations Assessment, Development and Evaluation (GRADE) was used to appraise the certainty of the evidence.[16] Evidence for each incidence measure was assessed. Evidence could be 'high certainty, 'moderate certainty, 'low certainty or 'very low certainty'. All ratings started at the 'moderate certainty' level as recommended for non-randomised study designs. Two authors (WD, MW) independently rated the evidence on the basis of study limitations, inconsistency, imprecision, and publication bias.

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Results

After removal of duplicates, a total of 451 articles were screened based on title and abstract. This revealed a total of 128 potentially eligible studies. After full text assessment of these 128 studies, thirteen studies met the inclusion criteria and were included in this review (Figure 1).[1,4,23,28,29,5,6,9,18–22]

Study characteristics

Table 1 presents the key characteristics of the included studies. The total number of included players among thirteen studies was more than 3868, with an overall exposure of 2,068,200 hours. All studies included men. Two studies evaluated acute hamstring injuries in women.

Study observation period

The start of the 'soccer season' was used in eleven studies[1,4,29,5,6,9,18–20,22,23] with different definitions of soccer season. Seven studies recorded injuries during the pre-season and usual soccer season;[5,9,18–20,22,23] four studies recorded injuries during competition only [1,4,6,29]. One study started a month before the start of the second half of the competition[21] (January); one started in the middle of the second half of the competition[28] (March).

The duration of data collection differed among the studies ranging from one[1,4,19,20,29] to eleven seasons[9]. The duration of a season varied from six[1] to twelve months[5,9,23]. One study did not define the duration of a soccer season.[6] The observation period for the two studies that started during the season, was one calendar year.[21,28]

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Study outcome definitions

Box 1 shows the definitions used. 'Acute hamstring injuries' was used as an outcome in 11 studies[1,5,6,9,18–21,23,2822], two studies[4,29] used 'acute injury to the posterior upper leg'. Ten studies used a 'time loss' injury definition [1,5,6,9,18–20,23,29], three studies used a 'tissue' injury definition[4,21,28]. In all studies 'medical staff' or a team physician were responsible for diagnosing hamstring injury.

Study quality

Overall, one of the studies[20] was awarded stars for all of the items of study quality (Table 2). Three studies[4,20,29] reported the injury status at baseline of players at risk of hamstring injuries. Three studies[18,20,28] provided a clear definition for an acute hamstring injury. All studies scored one or more stars on the three other items in domains 'selection' and 'other'.

Certainty of evidence (GRADE)

Certainty of the evidence was of very low level, with the exception of one subgroup rated 'low' and one subgroup rated 'moderate' (Table 3). The main reasons to rate down the certainty of the evidence were study limitations (n=11, 92%) and inconsistency (n=9, 75%). A subgroup was considered to have study limitations when the average of three stars or more were missing in the quality assessment. We downgraded for inconsistency when a subgroup had two or more points of heterogeneity. None of the subgroups were rated down for imprecision. We did not rate down for publication bias as there were fewer than 10 studies per incidence measure.

Hamstring injury incidence in soccer

The incidence of acute hamstring injuries was 0.3-1.9 injuries per 1000 exposure hours (Table 4).

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In studies using 'time loss' injury definition the incidence was 0.4-1.9 injuries per 1000 exposure hours. Studies using 'tissue' injury definition had an incidence of 0.3-1.0 injuries per 1000 exposure hours.

Professional and amateur players

The incidence of acute hamstring injury in professional players was 0.5-1.9 injuries per 1000 exposure hours with 'time loss' injury definition.[1,5,6,9,18,19,22,23,29] The incidence in 1 study with professional players[28] and a 'tissue' injury definition 0.3 (95% CI 0.3 to 0.5) injuries per 1000 exposure hours.

In amateur players, one study[20] using 'time loss' injury definition reported an incidence of 0.6 (95% CI 0.4 to 1.1) injuries per 1000 exposure hours. Two studies[4,21] used 'tissue' injury definition and reported an incidence of 0.5-1.0 injuries per 1000 exposure hours.

Women and men

In women,[19,23] the incidence of time-loss acute hamstring injury was 0.3-0.5 injuries per 1000 exposure hours (mean 0.5 per 1000 exposure hours). In men, the incidence of time-loss acute hamstring injury was 0.4-1.9 injuries per 1000 exposure hours.[1,5,6,9,18–20,23,29] The incidence of acute hamstring injuries using 'tissue' definition was 0.3-1.0 injuries per 1000 exposure hours[4,21,28].

Proportion of hamstring injuries from total injuries in soccer

Acute hamstring injuries accounted for 5% to 15% of all injuries in soccer (n = 7 studies, 9 populations; Table 1).[1,4,9,18,19,23,29].

Hamstring injury recurrence

Ten out of thirteen studies (77%) registered recurrent acute hamstring injuries (Table 4, Figure 2). Overall recurrence rate in the eight studies with available data ranged between 4% to 68%, with no limit in period of time. In seven studies, players were retrospectively asked about acute hamstring

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injuries (=index injury) in the previous year upon study entry. [1,4,19–21,28,29] In these studies, recurrent injury was defined as sustaining another hamstring injury in the same leg during the (prospective) study period of 6 to 12 months. Subsequent acute hamstring injuries were labelled as 'recurrent'. Time categories of recurrent injuries were different among these studies. Two of these studies used the definition as described in the 2006 consensus[12] (box 1) and differentiated in early (<2 months) and late recurrence (2-12 months).[19,21] However, data on these recurrent injuries were not available in one study[19]. The other five studies with retrospective information on index injuries used their own definition of time categories of recurrent injuries. One study had differentiation in five time categories varying from a recurrence 'within 1 month' to 'beyond 12 months'.[1] Four studies had no limit of time for recurrence (see Figure 2).[4,20,28,29]

Three studies prospectively registered injuries as 'recurrent' when another acute hamstring injury in the same leg occurred during the prospective study period. [5,6,18] All of these studies used the definition as described in the 2006 consensus [12](box 1). One study[6] recorded early (<2 months) recurrences, one study[18] had two groups with early and late (2-12 months) recurrences and one study[5] differentiated between early, late and delayed (>12 months) recurrence. Data were unavailable in one study with registration of early and late recurrent injuries.[18] Overall recurrence ranged from 17% to 34%. [5,6] Early recurrences ranged from 8% to 17%.

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Discussion

We evaluated the incidence, the relative percentage and recurrence rate of acute hamstring injuries in adult soccer. The incidence of acute hamstring injuries across more than 3868 players in thirteen included studies ranged from 0.3 to 1.9 injuries per 1000 exposure hours. In women, the incidence was 0.3-0.5 injuries per 1000 exposure hours;[19,23] in men the incidence was 0.3-1.9 injuries per 1000 exposure hours. [1,4,28,29,5,6,9,18–21,23] Acute hamstring injuries account for 5 to 15% of all injuries in soccer.[1,4,9,18,19,23,29] The recurrence rate of hamstring injuries was 4% to 68%. These data were heterogenous and should be interpreted with caution.

Possible explanations for the substantial heterogeneity in hamstring injury incidence are variations in injury definitions, diverse approaches for observing and registering hamstring injuries, and methodological shortcomings in the approach to injury surveillance. We rated down the certainty of the evidence to very low for most incidence measures on the basis of study limitations and inconsistency.

Shortcomings in the field of research on hamstring injuries

We noted limitations in the planning, conduct and reporting of hamstring studies. We highlight the most pressing issues here, and provide guidance for future studies on how to conduct better research, and why this is important.

- *Lack of definition of hamstring injury*

What constitutes an acute hamstring injury is rarely reported. Only three[12,20,28] out of thirteen studies clearly and comprehensively defined all aspects of the acute hamstring injury. Most of the studies lacked a description of the character of commencement of the hamstring injury, acute or non-acute. Acute hamstring injury[27] and the non-acute overuse hamstring injury[14] are two different types of injuries with a different etiology, presentation and prognosis. We included acute hamstring injuries only. Authors were contacted when the character of the hamstring injury was unclear; we were able to

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include four more studies, resulting in thirteen inclusions.[6,9,19,21]. We did not award a star on item 6; 'case definition' for these studies during quality assessment.

- *Variation of injury definition in the included studies*

Injury definitions varied across studies, with 'time loss' used in ten studies'[1,5,6,9,18–20,23,29] and 'tissue' injury' in 3 studies[4,21,28]. 'Tissue injury is a physical complaint, while 'time loss' injury definition is only registered when this complaint leads to time loss in play. Using 'tissue' injury definition could lead to over-registration of the number of hamstring injuries. Minor soreness not affecting the ability to play counts as an injury. However, without registration of 'tissue' injury definition, the burden of injury can be underestimated. Being able to play with an injury does not equal being uninjured. Injury incidence for the studies using the definition 'tissue' injury was within the range of incidence for the included studies using the definition of 'time loss' injury.

- *Variation in registration of recurrent injuries among included studies*

Methods for registering recurrent acute hamstring injuries was a source of variation in the design and conduct of studies in the field. To reliably register the incidence of hamstring injuries, it is important to differentiate first-time acute hamstring injuries from recurrent injuries. Recurrent injury registration should be collected prospectively, following definitions from the consensus statement on collecting data in soccer.[12] Injuries sustained before the start of a study should be considered when identifying an injury as a recurrence.[12]

Ten [1,4–6,18–21,28,29] out of thirteen studies (77%) registered recurrent injuries. Information on injuries before start of the study was used in seven studies[1,4,19–21,28,29] to identify recurrent injuries. However, bias may be introduced when these data are collected retrospectively, especially when these data are obtained from the players without medical injury registration. Three studies[5,6,18] started scoring recurrences after an index injury during the study. Hence, first-time injuries in this study may be a mix index injuries and recurrent injuries.

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The lack of registration of injuries occurring prior to the trial period, may overestimate first-time injuries and underestimate recurrences. Thus, both methods may introduce bias. Future studies must employ measures to ensure homogeneity in definition and methodology in studies across the field of acute hamstring injuries. Studies in the field can then be compared and pooled in a meta-analysis.

Another source of heterogenicity was the duration of registration of recurrent injuries. One study[6] defined a recurrent injury only when this hamstring injury occurred within two months after the index injury, the early recurrence. This will underestimate the recurrence rate and overestimate the primary acute hamstring injury. Others[4,5,18–21,28,29] scored recurrent injuries for a longer period, including late (2-12 months) and delayed recurrences (beyond twelve months). Besides a more accurate incidence rate on primary and recurrent hamstring injuries, registration in these specific groups will give valuable information for risk at different points in times for return to sport.

Strengths and Limitations

We used a sensitive search strategy without restriction in language or year of publication and well-defined inclusion criteria. A large proportion of the literature on acute hamstring injuries has been excluded. Studies were excluded for not reporting exposure hours, expressing exposure in 'Athlete Exposure' and clustering injuries according to body part (i.e. 'upper leg' or 'thigh') without specification to the muscle group (i.e. 'hamstring' or 'posterior thigh'). Whenever 'hamstring injury' was not specified to acute or when non-acute or recurrent or data were missing, the author was contacted by mail to obtain the correct numbers on exposure and the acute hamstring injury. If studies did not fulfill the eligibility criteria, or when no information on exposures and acute hamstring injuries could be obtained, studies were excluded. Including these data would have provided a challenge to the interpretation of the incidence of acute hamstring injuries, the extrapolation of results to other populations and interstudy comparison.

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Study quality assessment was adjusted to the observational research questions of this review, using a modified Newcastle Ottawa Scale. Adjustments focused on methodologic quality, making this tool a more appropriate quality assessment for this specific review compared to the original version.

We contacted the lead author of one of the studies[21] to obtain data on exposure hours. With these data, we calculated injury incidence for acute hamstring injuries. For our epidemiological approach, we excluded players with missing exposure data from the analysis, whereas in the original trial, exposure hours were analysed on the basis of the intention-to-treat principle. Therefore, the incidence we calculated for this study was slightly lower than reported in the original report (0.5 vs 0.8, respectively).

A secondary research question was added after the protocol of the study appeared online, on the incidence of recurrent acute hamstring injuries. No other changes occurred.

Our review is limited by the number of studies (N=13) and heterogeneous methods across original studies, hampering a quantitative analysis of the incidence of acute hamstring injuries in soccer. This is an example that underpins the importance of common methodologies and clear definitions. In attempt to raise the inclusion rate, authors of eligible studies with missing data, e.g. exposure hours, were contacted by email. With a low response rate and lack of registration of exposure hours, the majority of these studies were excluded.

We excluded retrospective studies, limiting the amount of data. Excluding retrospective designs minimizes risk of bias, and it adheres to a consensus statement for the prospective data collection in soccer.[12]

Unpublished data on injury surveillance could be missing. Professional teams and leagues often register injury data without the intention to publish these data. It is unknown how many teams keep these registrations, and we have not searched for these data (grey literature).

Five recommendations for future research

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Observations from this systematic review stress the importance of a standardized approach to assessing the incidence of acute hamstring injuries.

Recommendation 1: Clear definition of all aspects of the studied injury, such as character (acute or non-acute) should be the foundation of thorough research.

Recommendation 2: A standardized approach is essential for registering exposure hours. We excluded studies using Athlete Exposure. This incidence measure counts individual sessions as opposed to accounting for the true exposure. For example, one session of training is '1' Athlete Exposure, however this could be 0.5 or 2 exposure hours. To facilitate comparison and review of literature, it is essential that the field uses the same methodologies, including recording exposure in minutes.[12]

Recommendation 3: Document exposure hours and specify injuries to the muscle group level. A large number of hamstring injuries are registered using the generic term 'thigh' injuries. With other thigh injuries, like quadriceps injuries, these data are useless for research on hamstring injuries. We excluded 51 studies because of missing data on exposure hours and the absence of specific numbers for hamstring injuries.

Recommendation 4: Intensity of play should be taken into account when expressing incidence of acute hamstring injuries. The validity of expressing exposure in 'hours' for the incidence of acute hamstring injuries has been questioned recently.[18] Acute hamstring injuries occur with rapid eccentric loading and might be less related to amount of exposure hours and more to intensity of play. [21] Using modern registration methods, these activities can be monitored. Many different electronic tracking systems are available (e.g. GPS trackers, motion sensors and video analytics). Technology is rapidly evolving and all of these have different strengths and limitations, with no current gold standard.

Recommendation 5: Acute hamstring injuries in female soccer players warrant further investigation. Only two of the included studies included female players. With the growing popularity of soccer among women[11], this is a field worthy of study.

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Conclusions

The incidence rate of acute hamstring injuries in training and matches ranged from 0.3 to 1.9 injuries per 1000 exposure hours. Acute hamstring injuries accounted for 5% to 13% of all injuries in soccer. The rate of hamstring injury recurrence was 4% to 68% and should be cautiously interpreted.

The strength of the evidence was limited by a lack of methodological rigor, the use of varying definitions for acute hamstring injuries, and a heterogeneous conduct of studies on acute hamstring injuries.

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Conflicts of interest, funding and ethical approval

The authors report no conflicts of interest. No funds or sources of financial assistance were used to conduct this review, or used to assist with the preparation of the manuscript. As a systematic review, ethical approval was not sought in line with our institutional guidance.

Key Points

Findings

The incidence of acute hamstring injuries in training and matches was 0.3 to 1.9 injuries per 1000 exposure hours. Acute hamstring injuries accounted for 5% to 13% of all soccer injuries.

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Implications

Knowledge of the incidence of acute hamstring injuries in soccer contributes to quantifying its burden and may facilitate the prevention of hamstring injuries by increasing awareness with players, clubs and coaches.

Caution

Certainty of the evidence ranged from moderate to very low. A lack of methodological rigor, the use of varying definitions for acute hamstring injuries, and a heterogeneous conduct of studies of acute hamstring injuries contributes to uncertainty.

Study details

Patient involvement

There was no patient involvement in this research.

Data sharing

All data relevant to the study are included in the article or are available as supplementary files.

Author contributions

All authors contributed to the review process, writing the manuscript and final adaptations

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References

1. Arnason A, Sigurdsson S, Gudmundsson A, Holme I, Engebretsen L, Bahr R. Risk Factors for Injuries in Football. *Am J Sports Med*. 2004;(June 2014). doi:10.1177/0363546503258912.
2. Bauer R, Steiner M, Kisser R, Macey SM, Thayer D. Unfälle in der EU: Ergebnisse des EuroSafe-Reports. *Bundesgesundheitsblatt - Gesundheitsforsch - Gesundheitsschutz*. 2014;57(6):673-680. doi:10.1007/s00103-014-1969-5.
3. Van Beijsterveldt AM, Steffen K, Stubbe JH, Frederiks JE, Van De Port IGL, Backx FJG. Soccer injuries and recovery in dutch male amateur soccer players: Results of a prospective cohort study. *Clin J Sport Med*. 2014;24(4):337-342. doi:10.1097/JSM.0000000000000028.
4. van Beijsterveldt AMC, van de Port IGL, Krist MR, et al. Effectiveness of an injury prevention programme for adult male amateur soccer players: a cluster-randomised controlled trial. *Br J Sports Med*. 2012;46(16):1114-1119. doi:10.1136/bjsports-2012-091277.
5. Carling C, Le Gall F, Orhant E. A four-season prospective study of muscle strain reoccurrences in a professional football club. *Res Sports Med*. 2011;19(2):92-102. doi:10.1080/15438627.2011.556494.
6. Carling C, McCall A, Le Gall F, Dupont G. The impact of short periods of match congestion on injury risk and patterns in an elite football club. *Br J Sports Med*. 2016;50(12):764-768. doi:10.1136/bjsports-2015-095501.
7. Donnelly JE, Hillman CH, Castelli D, et al. Physical Activity, Fitness, Cognitive Function, and Academic Achievement in Children. *Med Sci Sport Exerc*. 2016;48(6):1197-1222. doi:10.1249/MSS.0000000000000901.
8. Eime RM, Young JA, Harvey JT, Charity MJ, Payne WR. A systematic review of the psychological and social benefits of participation in sport for children and adolescents: Informing development of a conceptual model of health through sport. *Int J Behav Nutr Phys Act*. 2013;10. doi:10.1186/1479-5868-10-98.
9. Ekstrand J, Hägglund M, Kristenson K, Magnusson H, Waldén M. Fewer ligament injuries but no preventive effect on muscle injuries and severe injuries: an 11-year follow-up of the UEFA

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Champions League injury study. *Br J Sports Med*. 2013;47(12):732-737. doi:10.1136/bjsports-2013-092394.

10. Ekstrand J, Krutsch W, Spreco A, et al. Time before return to play for the most common injuries in professional football: a 16-year follow-up of the UEFA Elite Club Injury Study. *Br J Sport Med*. 2019. doi:10.1136/bjsports-2019-100666.

11. Fédération Internationale de Football Association (FIFA). FIFA Big Count 2006: 270 million people active in football. https://www.fifa.com/mm/document/fifafacts/bcoffsurv/bigcount.statspackage_7024.pdf. Published 2007.

12. Fuller CW, Ekstrand J, Junge A, et al. Consensus statement on injury definitions and data collection procedures in studies of football (soccer) injuries. *Br J Sports Med*. 2006;40:193-201. doi:10.1136/bjsm.2005.025270.

13. Garber CE, Blissmer B, Deschenes MR, et al. Quantity and quality of exercise for developing and maintaining cardiorespiratory, musculoskeletal, and neuromotor fitness in apparently healthy adults: Guidance for prescribing exercise. *Med Sci Sports Exerc*. 2011;43(7):1334-1359. doi:10.1249/MSS.0b013e318213fefb.

14. Goom TSH, Malliaras P, Reiman MP, Purdam CR. Proximal hamstring tendinopathy: Clinical aspects of assessment and management. *J Orthop Sports Phys Ther*. 2016;46(6):483-493. doi:10.2519/jospt.2016.5986.

15. Green S, Higgins JP, Green S. Cochrane Handbook. 2011. doi:http://dx.doi.org/10.1016/S1873-0418(01)80002-6.

16. Guyatt GH, Oxman AD, Vist GE, et al. GRADE: an emerging consensus on rating quality of evidence and strength of recommendations. *BMJ*. 2008;336(7650):924-926. doi:10.1136/bmj.39489.470347.AD.

17. Hägglund M, Waldén M, Bahr R, Ekstrand J. Methods for epidemiological study of injuries to professional football players: developing the UEFA model. *Br J Sports Med*. 2005;39(6):340-346. doi:10.1136/bjsm.2005.018267.

18. Hägglund M, Waldén M, Ekstrand J. Previous injury as a risk factor for injury in elite football:

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a prospective study over two consecutive seasons. *Br J Sports Med*. 2006;40(4):767-772. doi:10.1136/bjsm.2006.026609.

19. Hägglund M, Waldén M, Ekstrand J. zInjuries among male and female elite football players. *Scand J Med Sci Sport*. 2009;19(6):819-827. doi:10.1111/j.1600-0838.2008.00861.x.

20. Van De Hoef PA, Brink MS, Huisstede BMA, et al. Does a bounding exercise program prevent hamstring injuries in adult male soccer players? – A cluster-RCT. *Scand J Med Sci Sports*. 2019;29(4):515-523. doi:10.1111/sms.13353.

21. van der Horst N, Smits D-W, Petersen J, et al. The Preventive Effect of the Nordic Hamstring Exercise on Hamstring Injuries in Amateur Soccer Players: A Randomized Controlled Trial. *Am J Sports Med*. 2015;43(6):1-8. doi:10.1177/0363546515574057.

22. Kristenson K, Bjørneboe J, Waldén M, Andersen TE, Ekstrand J, Hägglund M. The Nordic Football Injury Audit : higher injury rates for professional football clubs with third-generation arti fi cial turf at their home venue. *Br J Sport Med*. 2013;(June). doi:10.1136/bjsports-2013-092266.

23. Larruskain J, Lekue JA, Diaz N, Odriozola A, Gil SM. A comparison of injuries in elite male and female football players: A five-season prospective study. *Scand J Med Sci Sports*. 2018;28(1):237-245.

24. Liu M, Wu L, Ming Q. How does physical activity intervention improve self-esteem and self-concept in children and adolescents? Evidence from a meta-analysis. *PLoS One*. 2015;10(8):1-17. doi:10.1371/journal.pone.0134804.

25. Moher D, Liberati A, Tetzlaff J, Altman DG, Prisma Group. Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement (Reprinted from Annals of Internal Medicine). *Ann Intern Med*. 2009;151(4):264-269. doi:10.1371/journal.pmed.1000097.

26. Mueller-Wohlfahrt HW, Haensel L, Mithoefer K, et al. Terminology and classification of muscle injuries in sport: The Munich consensus statement. *Br J Sports Med*. 2013;47(6):342-350. doi:10.1136/bjsports-2012-091448.

27. Pas HIMFL, Reurink G, Tol JL, Weir A, Winters M, Moen MH. Efficacy of rehabilitation (lengthening) exercises, platelet-rich plasma injections, and other conservative interventions in

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acute hamstring injuries: An updated systematic review and meta-analysis. *Br J Sports Med*. 2015;49(18):1197-1205. doi:10.1136/bjsports-2015-094879.

28. Petersen J, Thorborg K, Nielsen MB, Hölmich P. Acute hamstring injuries in Danish elite football: A 12-month prospective registration study among 374 players. *Scand J Med Sci Sports*. 2010;20(4):588-592. doi:10.1111/j.1600-0838.2009.00995.x.

29. Stubbe JH, Van Beijsterveldt AMMC, Van Der Knaap S, et al. Injuries in professional male soccer players in the Netherlands: A prospective cohort study. *J Athl Train*. 2015;50(2):211-216. doi:10.4085/1062-6050-49.3.64.

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BOX 1: Definitions used from 2005 UEFA model by Hagglund et al[17] and 2006 Consensus Fuller et al[12]

Injury definition

'Tissue' injury

Any physical complaint sustained by a player that results from a soccer match or soccer training, irrespective of the need for medical attention or time loss from soccer activities.

'Medical attention' injury

An injury that results in a player receiving medical attention.

'Time loss' injury

An injury that results in a player being unable to take a full part in future soccer training or match play.

Recurrent injury

An injury of the same type and at the same site as an index injury and which occurs after a player's return to full participation from the index injury.

'early recurrence'

A recurrent injury occurring within two months of a player's return to full participation

'late recurrence'

A recurrent injury occurring 2–12 months after a player's return to full participation

'delayed recurrence'

A recurrent injury occurring more than 12 months after a player's return to full participation.

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TABLE 1. Key characteristics of included studies

Author (year)	Study Design	Number of included players (N=)	Level	Gender	Age (Mean ± SD)	Type injury definition	Start study	Observation duration (months)
Arnason et al (2004)[1]	Cohort	306	Elite	M	24	Time loss	May	6
Carling et al (2011)[5]	Cohort	46	Prof	M	17–34*	Time loss	June	12-48
Carling et al (2016)[6]	Cohort	14	Prof	M	NR	Time loss	NR	NR (6 seasons)
Ekstrand et al (2013)[9]	Cohort	1743	Prof	M	NR	Time loss	July	12-132
Hägglund et al (2006)[18]	Cohort	263	Semi-prof	M	25 ± 5	Time loss	January	2x 11
Hägglund et al (2009)[19]	Cohort	Male 239 Female 228	M: Prof F: Elite	M/F	Male 25 ± 5 Female 23 ± 4	Time loss	January	10
Hoef et al (2019)[20]	RCT	Control 171	Amateur	M	22.2 ± 3.1	Time loss	September	9
Kristenson et al (2013)[22]	Cohort	NR	Prof	M	Turf 25.2 ± 5.0 Grass 25.0 ± 4.9	Time loss	January	2x11
Larruskain et al (2018)[23]	Cohort	Male 50 Female 35	Prof	M/F	Male 25 ± 4 Female 25 ± 5	Time loss	July	60
Petersen et al (2010)[28]	Cohort	242	Prof	M	17-35 and older*	Tissue	March	12
Stubbe et al (2015)[29]	Cohort	217	Prof	M	24.6 ± 4.3	Time loss	August	9
Van Beijsterveldt et al (2012)[4]	RCT	Control 233	Amateur	M	25.1 ± 4.3	Tissue	September	9
Van der Horst et al (2015)[21]	RCT	Control 81†	Amateur	M	24.6 ± 4.1	Tissue	January	12

Abbreviations: RCT = Randomised Controlled Trial, Prof = Professional, M = Male, F = Female, SD = Standard deviation, NR= Not reported

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*Range of age; mean and SD not reported

† Number of players for which complete exposure and injury data was available. Total control group, n = 287.

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TABLE 2. Quality assessment

Author (year)	Item *						
	1: ***Population description***	2: ***Population recruitment***	3: ***Injury status at baseline***	4: ***Surveillance methods***	5: ***Time of observation***	6: ***Case definition***	7: **Other methodological aspects**
Arnason et al (2004)[1]	+	-	-	+	+	-	+
Carling et al (2011)[5]	+	+	-	+	+	-	+
Carling et al (2016)[6]	-	+	-	+	+	-	-
Ekstrand et al (2013)[9]	-	+	-	+	+	-	+
Hägglund et al (2006)[18]	+	+	-	+	+	+	+
Hägglund et al (2009)[19]	+	+	-	+	+	-	+
Hoef et al (2019)[20]	+	+	+	+	+	+	+
Kristenson et al (2013)[22]	-	-	-	+	+	-	-
Larruskain et al (2018)[23]	+	-	-	+	+	-	+
Petersen et al (2010)[28]	+	-	-	+	+	+	+
Stubbe et al (2015)[29]	+	-	+	+	+	-	-
Van Beijsterveldt et al (2012)[4]	+	-	+	+	+	-	-
Van der Horst et al (2015)[21]	+	+	-	+	+	-	+

*: Please see all items in appendix 2; Modified Newcastle Ottawa Scale.
+ marks a star of quality that is present
- marks a star of quality that is absent

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TABLE 3. **Certainty of the evidence (GRADE)**

Incidence measure per subgroup	**Hamstring incidence (per 1000 exposure hours)**	**Quality of evidence**	**Downgrading due to**
Total of all subgroups	0.3-1.9	Very low	Study limitations*, inconsistency†
Injury definition – all groups			
Time loss' injury definition	0.4-1.9	Very low	Study limitations*, inconsistency†
Tissue' injury definition	0.3-1.0	Very low	Study limitations*, inconsistency†
Professional players			
'Time loss' injury definition	0.5-1.9	Very low	Study limitations*, inconsistency†
'Tissue' injury definition	0.3 (95% CI 0.3-0.5)	Moderate	None
Amateur players			
'Time loss' injury definition	0.6 (95% CI 0.4-1.1)	Very low	Study limitations*
'Tissue' injury definition	0.5-1.00	Very low	Study limitations*, inconsistency†
Men			
'Time loss' injury definition	0.4-1.9	Very low	Study limitations*, inconsistency†
'Tissue' injury definition	0.3-1.0	Very low	Study limitations*, inconsistency†
Women			
'Time loss' injury definition	0.3-0.5	Low	Study limitations*
Other	**Percentage**		
Hamstring/total injuries	5.0 % - 15.4 %	Very low	Study limitations*, inconsistency†
Recurrence rate	4.2 % - 67.7 %	Very low	Study limitations*, inconsistency†

95% CI = 95% confidence interval
* Study limitations: serious. Three or more missing stars in the quality assessment
† Inconsistency: serious. Two or more points of heterogeneity

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TABLE 4. Total numbers of hamstring injuries, exposure (match/training) and injury incidence.

Author (year)	Number of acute hamstring injuries (n=)	Total number of injuries (n=)	Total Exposure (h)	Match exposure (h)	Training exposure (h)	Match/ total exposure (%)	Hamstring incidence (hamstring injuries/ 1000h)	Total injury incidence rate (total injuries/ 1000h)	Percentage hamstring / total injuries (%)	Time of recurrence (months)	Number of recurrent hamstring injuries (n=)	Percentage recurrent hamstring injuries per timeframe (%)	Total percentage of recurrent hamstring injuries (%)
Arnason et al (2004)[1]	10 ‡	244	33839	5968	27871	17.6	0.9		12.7	<1 1-3 3-6 6-12 >12 Unknown‡	5 †‡ 3 †‡ 2 †‡ 2 †‡ 5 †‡ 4 †‡	33.3 †‡ 23.1 †‡ 16.7 †‡ 16.7 †‡ 33.3 †‡ 28.6 †‡	67.7 †‡
Carling et al (2011)[5]	33	-	33432	3548	29884	10.6	1.0	-	-	<2 2-12 >12	3 *‡ 8 *‡ 6 *‡	8.3 *‡ 19.5 *‡ 15.4 *‡	34 *‡
Carling et al (2016)[6]	90‡	-	47468‡	6623‡	40845‡	16.2	1.9	-	-	<2	19 *‡	17.4 *‡	17.4 *‡
Ekstrand et al (2013)[9]	1025	8029	1057201	168952	888249	16.0	1.0	7.6	12.8	-	-	-	-
Hägglund et al (2006)[18]	81‡	1189	155967	20145	135822	12.9	0.5	7.6	6.8	<2 >2	- *§	- *§	- *§
Hägglund et al (2009)[19]	Male 36‡ Female 29‡	Male 548 Female 299	Male 71361 Female 54156	Male 9064 Female 6787	Male 62315 Female 46469	Male 12.7 Female 12.5	Male 0.5 Female 0.5	Male 7.7 Female 5.5	Male 6.6 Female 10.0	<2 >2	- †§	- †§	- †§

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Hoef et al (2019)[20]	14	-	21717	6276	15441	28.9	0.6	-	-	No limit	4 †	-	22 †
Kristenson et al (2013)[22]	161	- ‖	367490	48922	318568	-	0.4	- ‖	- ‖	-	-	-	-
Larruskain et al (2018)[23]	Male 18 ‡ Female 8 ‡	Male 323 Female 160	Male 38878 Female 25394	Male 5391 Female 3544	Male 33487 Female 21850	Male 13.9 Female 14.0	Male 0.5 Female 0.3	Male 8.3 Female 6.3	Male 5.6 Female 5.0	-	-	-	-
Petersen et al (2010)[28]	46	-	134011	15367	118644	11.5	0.3	-	-	No limit	8 †	14.8 †	14.8 †
Stubbe et al (2015)[29]	23	286	46194	5182	41012	11.2	0.5	6.2	15.4	No limit	1	4.2	4.2
Van Beijsterveldt et al (2012)[4]	23	220	22647‡	6342‡	16305‡	28	1.0	9.7	10.5	No limit	6 †‡	20.7 †‡	20.7 †‡
Van der Horst et al (2015)[21]	81‡	4‡	7899‡	2957‡	4942‡	62.6	0.5	-	-	<2 >2	0 †‡ 2 †‡	0 †‡ 30 †‡	30 †‡

Abbreviations: h = hour, n=number

* Only recurrent injuries after a first injury in study was counted
† Information before start of study was used to score recurrent injuries
‡ Information not in article, retrieved from author
§ Data was not available
‖ Conflicting data

FIGURE 1. PRISMA Study selection flow chart

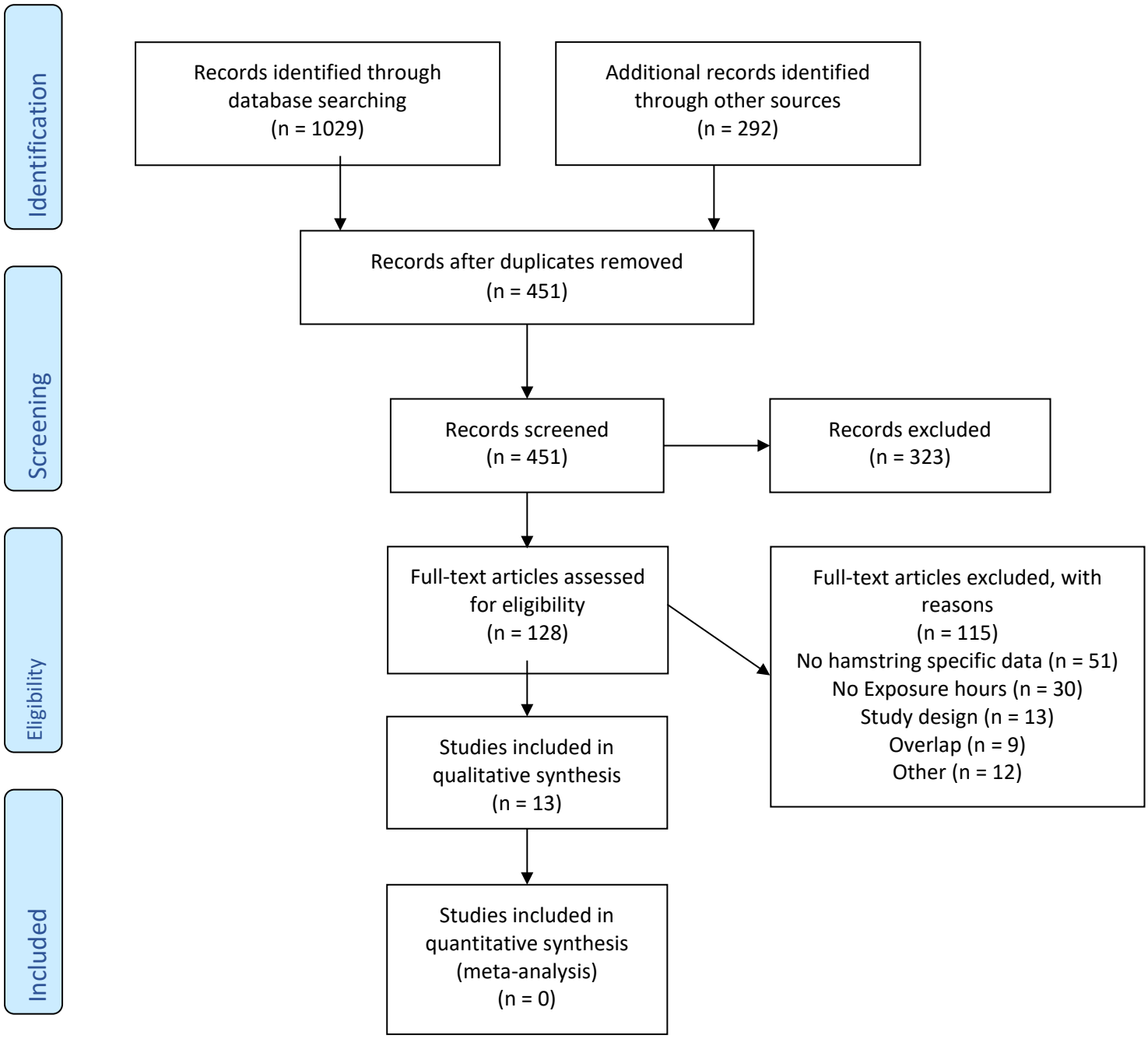

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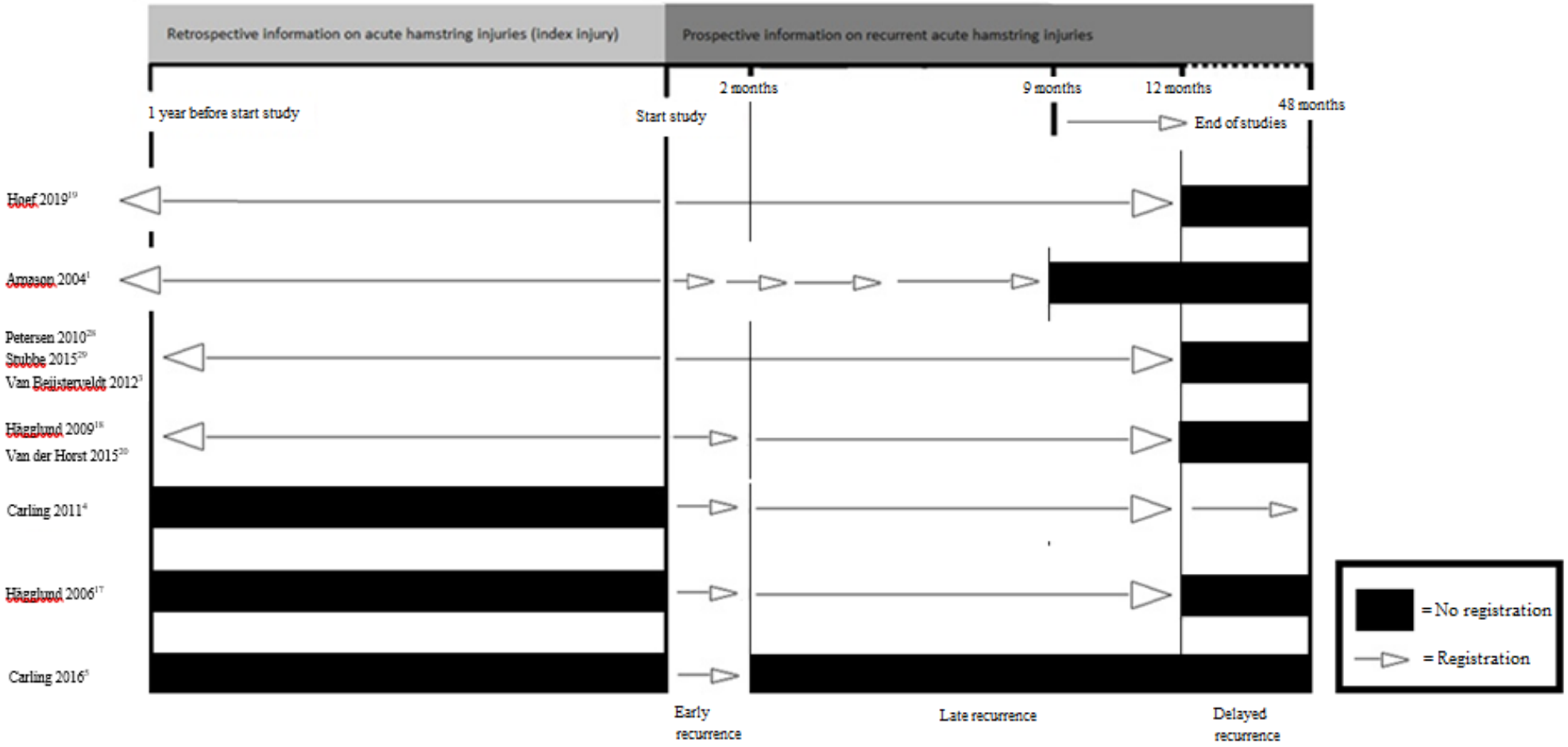
FIGURE 2. Time line of recording recurrent injuries in ten studies.[1,4–6,18–21,28,29]



SEARCH STRATEGY

Pubmed:
((("Thigh"[Mesh] OR hamstring*[tiab] OR thigh*[tiab] OR semitendino*[tiab] OR semimembrano*[tiab] OR biceps femor*[tiab])) AND ("Wounds and Injuries"[Mesh:NoExp] OR "Rupture"[Mesh:NoExp] OR "Sprains and Strains"[Mesh:NoExp] OR injur*[tiab] OR ruptur*[tiab] OR tear*[tiab] OR avuls*[tiab] OR strain*[tiab] OR sprain*[tiab])) AND ("Epidemiology"[Mesh] OR epidemiol*[tiab] OR "Incidence"[Mesh] OR inciden*[tiab] OR occur*[tiab])) AND ("Soccer"[Mesh] OR soccer*[tiab] OR football*[tiab])

Embase:
(exp thigh/ OR (thigh* or semitendino* or semimembrano* or biceps femor* or gracilis).ti,ab) AND (exp injury/ OR exp rupture/ OR exp muscle strain/ OR (injur* or ruptur* or tear* or avuls* or strain* or sprain*).ti,ab) AND (exp incidence/ OR exp epidemiology/ OR (epidemiol* or inciden* or occur*).ti,ab) AND (exp soccer/ OR (soccer* or football*).ti,ab.)

CINAHL:
((MH “Thigh”) OR (TI (hamstring* OR thigh* OR semitendino* OR semimenbrano* OR biceps femor*)) OR (AB (hamstring* OR thigh* OR semitendino* OR semimenbrano* OR biceps femor*)) AND ((MH “Wounds and Injuries”) OR (MH “Rupture”) OR (MH “Sprains and Strains”)) OR (TI (injur* OR ruptur* OR tear* OR avuls* OR strain* OR sprain*)) OR (AB (injur* OR ruptur* OR tear* OR avuls* OR strain* OR sprain*)) AND ((MH “Incidence”) OR (MH “Epidemiology”)) OR (TI (epidemiol* OR inciden* OR occur*)) OR (AB(epidemiol* OR inciden* OR occur*)) AND (MH “Soccer”) OR (TI (soccer* OR football(MH “Soccer”) OR (TI (soccer* OR football*)) OR (AB (soccer* OR football*)) *)) OR (AB (soccer* OR football*))

Sportdiscus:
(TI (hamstring* OR thigh* OR semitendino* OR semimenbrano* OR biceps femor*) OR AB (hamstring* OR thigh* OR semitendino* OR semibrembrano* OR biceps femor*)) AND (TI (injur* OR ruptur* OR tear* OR avuls* OR strain* OR sprain*) OR AB (injur* OR ruptur* OR tear* OR avuls* OR strain* OR sprain*)) AND (TI (epidemiol* OR inciden* OR occur*) OR AB (epidemiol* OR inciden* OR occur*)) AND (TI (soccer* OR football*) OR AB (soccer* OR football*))

Cochrane:
(MeSH descriptor: [Thigh] explode all trees OR (hamstring* or thigh* or semitendino* or semimembrano* or biceps femor*) AND (MeSH descriptor: [Wounds and Injuries] this term only OR MeSH descriptor: [Rupture] this term only OR MeSH descriptor: [Sprains and Strains] this term only OR (injur* or ruptur* or tear* or avuls* or strain* or sprain*)) AND (MeSH descriptor: [Epidemiology] explode all trees OR MeSH descriptor: [Incidence] explode all trees OR (epidemiol* or inciden* or occur*)) AND (MeSH descriptor: [Soccer] explode all trees OR MeSH descriptor: [Football] explode all trees OR (soccer* or football*))

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Modified Newcastle Ottawa Scale for incidence acute hamstring injury review[Wells GA, Shea B, O'Connell D et al. The Newcastle-Ottawa Scale (NOS) for assessing the quality of nonrandomised studies in meta-analyses. http://www.ohri.ca/programs/clinical_epidemiology/oxford.asp]

Selection

#1: Population description: One star was awarded when the population at risk was described (i.e. competition level, gender, age)

#2: Population recruitment: One star was awarded when it was described how the population under study was arrived at, and, the entire population participated or a random sampling (fraction) method was used to follow a sample of the population at risk for acute hamstring injury.

#3: Injury status at baseline: One star was awarded when the health and injury status at baseline was described.

Observation

#4: Surveillance methods: One star was awarded when it was stated how the incidence of acute hamstring injuries were monitored over time, by an active or passive surveillance method.

#5: Time of observation: One star was awarded when the time of observation was stated.

Outcome (incidence):

#6: Case definition: One star was awarded when the study defined "acute hamstring injury" or "acute injury to the posterior thigh".

#7: Other methodological aspects*

One star was awarded when all other methods were found appropriate.

* e.g. attrition aspect (in case of cumulative incidence/exposure-rate incidence measures): > 10% lost to follow up? No star was provided.

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